A Finite 3-Adic Obstruction to Odd Collatz Cycles

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Abstract

We study the odd-to-odd dynamics of the Collatz map via the associated exponent sequence defined by $3n + 1 = 2^{e(n)}m$. Finite windows of consecutive odd steps satisfy exact algebraic identities which induce compatibility constraints modulo powers of 3. Under the hypothesis of an odd Collatz cycle, these constraints must hold simultaneously for all circular windows of fixed length.

We encode these constraints as finite directed graphs. A complete enumeration of directed cycles in the graphs corresponding to window lengths $k = 3$ and $k = 4$ shows that no nonconstant exponent configuration satisfies the required 3-adic compatibility conditions. The remaining constant exponent case is then eliminated by a direct algebraic argument.

This yields a finite obstruction to the existence of nontrivial odd Collatz cycles.

1 Introduction

The Collatz map sends a positive integer n to $n/2$ if n is even and to $3n + 1$ if n is odd. Despite its elementary formulation, the global behavior of this iteration remains unresolved. The problem has been extensively studied in the literature; see for example [1, 2]. One of the fundamental open questions is whether nontrivial cycles exist.

It is natural to restrict attention to the induced odd-to-odd dynamics. Each odd step can be written uniquely in the form

$$3n + 1 = 2^{e(n)}m,$$

where m is odd and $e(n) = v_2(3n + 1)$. Iterating this representation encodes odd Collatz trajectories as sequences of exponents.

The central observation of this work is that finite windows of consecutive odd steps satisfy exact algebraic identities. When reduced modulo powers of 3, these identities impose compatibility constraints on ordered blocks of exponents that must hold simultaneously under all circular shifts along a hypothetical cycle.

These circular compatibility conditions are encoded as finite directed graphs. At window length three, the resulting 3-adic constraints force the exponent sequence to be constant modulo 18. At window length four, a refinement modulo 81 eliminates all nonconstant configurations. The remaining constant case is resolved by a direct algebraic argument.

The argument is entirely finite and local in nature. It relies only on exact window identities and exhaustive compatibility checks on finite graphs, without invoking global convergence assumptions.

2 Statement of the Main Result

Let $T\colon \mathbb{N} \to \mathbb{N}$ denote the Collatz map,

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n + 1 & \text{if } n \text{ is odd.} \end{cases}$$

We restrict attention to the induced odd-to-odd dynamics. For an odd integer n, define

$$3n + 1 = 2^{e(n)} m,$$

where m is odd and $e(n) = v_2(3n + 1)$.

An *odd Collatz cycle* is a finite sequence of odd integers

$$n_0, n_1, \dots, n_{p-1}$$

such that

$$3n_i + 1 = 2^{e_i} n_{i+1} \quad (i = 0, \dots, p - 1),$$

with indices taken modulo p.

The trivial cycle consists of the single fixed point $n_0 = 1$.

Theorem 2.1. *There exist no nontrivial odd Collatz cycles.*

3 Algebraic Window Identity

Let (n_i) be an odd Collatz trajectory and (e_i) its exponent sequence,

$$3n_i + 1 = 2^{e_i} n_{i+1}.$$

3.1 Iterated identity

Fix an integer $k \geq 1$. By repeated substitution, one obtains the exact identity

$$3^k n_{i+k} = 2^{E_i} n_i - \sum_{j=0}^{k-1} 3^j \, 2^{\,e_{i+j+1} + \cdots + e_{i+k-1}}, \tag{1}$$

where

$$E_i = e_i + e_{i+1} + \cdots + e_{i+k-1}.$$

3.2 Reduction modulo 3^k

Reducing (1) modulo 3^k eliminates the left-hand side and yields

$$2^{E_i} n_i \equiv \sum_{j=0}^{k-1} 3^j \, 2^{\,e_{i+j+1} + \cdots + e_{i+k-1}} \pmod{3^k}. \tag{2}$$

Since $\gcd(2, 3^k) = 1$, the factor 2^{E_i} is invertible modulo 3^k. Thus we may rewrite (2) in the form

$$n_i \equiv R_k(e_i, \dots, e_{i+k-1}) \pmod{3^k}, \tag{3}$$

where

$$R_k(e_i, \dots, e_{i+k-1}) = 2^{-E_i} \sum_{j=0}^{k-1} 3^j \, 2^{\,e_{i+j+1} + \cdots + e_{i+k-1}} \pmod{3^k}.$$

3.3 Consequence for cycles

If $(n_0, \ldots, n_{p-1})$ is an odd Collatz cycle, then for every $i \in \mathbb{Z}/p\mathbb{Z}$ the congruence (3) holds for the window

$$(e_i, e_{i+1}, \ldots, e_{i+k-1}),$$

with indices taken modulo p.

4 Circular Compatibility as a Finite Graph

Fix $k \geq 1$.

For each ordered k-tuple of exponents

$$(e_i, \ldots, e_{i+k-1}),$$

define the residue

$$R_k(e_i, \ldots, e_{i+k-1}) \in \mathbb{Z}/3^k\mathbb{Z}$$

as in (3).

4.1 Compatibility condition

Along an odd trajectory we have

$$3n_i + 1 = 2^{e_i} n_{i+1}.$$

Reducing modulo 3^k and substituting the window residues yields

$$R_k(e_{i+1}, \ldots, e_{i+k}) \equiv \big(3R_k(e_i, \ldots, e_{i+k-1}) + 1\big)\, 2^{-e_i} \pmod{3^k}. \tag{4}$$

4.2 Directed graph encoding

For each fixed window length k, we associate a directed graph G_k as follows:

- Vertices: all ordered k-tuples of exponent residues modulo the multiplicative order of 2 modulo 3^k.
- Directed edge:

  $$(e_0, \ldots, e_{k-1}) \longrightarrow (e_1, \ldots, e_k)$$

  whenever the compatibility condition (4) holds.

Lemma 4.1. *For each fixed k, if an odd Collatz cycle exists, then its length-k exponent windows form a directed cycle in the associated graph G_k.*

Proof. In a cycle of length p, every length-k window occurs, including those wrapping around the cycle. Each must satisfy (4). Therefore the sequence of windows produces a directed cycle in G_k. □

5 Finite Obstruction at Window Length $k = 3$

We specialize the window identity to length $k = 3$ and work modulo $3^3 = 27$. Since 2 is a unit modulo 27 and powers of 2 are periodic modulo 27, only exponent residues modulo the multiplicative order of 2 mod 27 are relevant. For $k \geq 1$, the multiplicative order of 2 modulo 3^k is

$$\mathrm{ord}_{3^k}(2) = 2 \cdot 3^{k-1}, \qquad \text{hence} \qquad \mathrm{ord}_{27}(2) = 18.$$

Therefore, every residue condition modulo 27 depends only on exponents reduced modulo 18.

Lemma 5.1. *For $k = 3$, the residue map modulo 27 depends only on exponent classes modulo 18. More precisely, the quantity*

$$R_3(a, b, c) = 2^{-(a+b+c)}\left(2^{b+c} + 3 \cdot 2^c + 9\right) \pmod{27}$$

is determined entirely by the classes of a, b, c in $\mathbb{Z}/18\mathbb{Z}$. Moreover, the compatibility condition

$$R_3(b, c, d) \equiv \big(3R_3(a, b, c) + 1\big)\, 2^{-a} \pmod{27}$$

also depends only on the classes of a, b, c, d modulo 18.

Proof. Since $\mathrm{ord}_{27}(2) = 18$, we have

$$2^{x+18m} \equiv 2^x \pmod{27} \qquad \text{for all } x, m \in \mathbb{Z}.$$

Because 2 is invertible modulo 27, the same holds for inverses:

$$2^{-(x+18m)} \equiv 2^{-x} \pmod{27}.$$

Thus every power of 2 and every inverse power of 2 appearing in the definition of $R_3(a, b, c)$ depends only on the corresponding exponent modulo 18. Therefore $R_3(a, b, c)$ itself depends only on (a, b, c) mod 18. The same argument applies to the compatibility condition, since both $R_3(a, b, c)$ and 2^{-a} depend only on the corresponding residue classes modulo 18. □

5.1 The length–three residue map

For a triple $(a, b, c) \in (\mathbb{Z}/18\mathbb{Z})^3$, define

$$R_3(a, b, c) := 2^{-(a+b+c)}\left(2^{b+c} + 3 \cdot 2^c + 9\right) \in \mathbb{Z}/27\mathbb{Z}, \tag{5}$$

where all powers of 2 and inverses are taken modulo 27. This is well-defined since 2 is invertible modulo 27.

If (e_i, e_{i+1}, e_{i+2}) is a realizable length–three window along an odd Collatz trajectory and we reduce exponents modulo 18, then the window identity implies

$$n_i \equiv R_3(e_i, e_{i+1}, e_{i+2}) \pmod{27}.$$

5.2 Circular compatibility as a directed graph

Along an odd trajectory we have

$$3n_i + 1 = 2^{e_i} n_{i+1}.$$

Reducing modulo 27 and substituting the window residues gives the compatibility constraint

$$R_3(e_{i+1}, e_{i+2}, e_{i+3}) \equiv \big(3\, R_3(e_i, e_{i+1}, e_{i+2}) + 1\big)\, 2^{-e_i} \pmod{27}, \tag{6}$$

with all e_j understood modulo 18.

Define a directed graph G_3 whose vertex set is $(\mathbb{Z}/18\mathbb{Z})^3$. We draw a directed edge

$$(a, b, c) \longrightarrow (b, c, d)$$

if and only if the compatibility congruence (6) holds.

Lemma 5.2 (Real cycles project to G_3)**.** *Let*

$$(n_0, n_1, \ldots, n_{p-1})$$

be an odd Collatz cycle with exponent sequence (e_i). Then the sequence of triples

$$(e_i, e_{i+1}, e_{i+2}) \bmod 18$$

forms a directed cycle in G_3.

Proof. For every index i along the cycle, the length–three window identity implies

$$n_i \equiv R_3(e_i, e_{i+1}, e_{i+2}) \pmod{27}.$$

Since the window identity is an exact algebraic identity for every odd Collatz trajectory, this congruence holds for every index i along the cycle.

The odd-to-odd recurrence gives

$$3n_i + 1 = 2^{e_i} n_{i+1}.$$

Reducing modulo 27 yields

$$n_{i+1} \equiv (3n_i + 1)\, 2^{-e_i} \pmod{27}.$$

Substituting the window residues gives

$$R_3(e_{i+1}, e_{i+2}, e_{i+3}) \equiv \big(3R_3(e_i, e_{i+1}, e_{i+2}) + 1\big)\, 2^{-e_i} \pmod{27}.$$

By Lemma 5.1, this condition depends only on the residue classes modulo 18. Therefore the triples

$$(e_i, e_{i+1}, e_{i+2}) \bmod 18$$

follow the edge rule defining G_3.

Since the cycle is circular, these triples form a directed cycle in G_3. □

5.3 Consequence: rigidity modulo 18

The key finite obstruction at window length three is the following.

Lemma 5.3. *The directed graph G_3 has no directed cycles other than the constant triples modulo* 18

$$(a, a, a) \to (a, a, a), \qquad a \in \mathbb{Z}/18\mathbb{Z}.$$

Recall that the vertices of G_3 are triples of exponent residues

$$(a, b, c) \in (\mathbb{Z}/18\mathbb{Z})^3.$$

Thus a vertex (a, a, a) represents three consecutive exponents that are equal modulo 18, not necessarily equal as integers.

Lemma 5.3 is a finite statement about a directed graph with 18^3 vertices. Once it holds, any exponent sequence arising from a hypothetical odd Collatz cycle must satisfy

$$e_i \equiv e_{i+1} \equiv \cdots \equiv e_{i+p-1} \pmod{18}.$$

Indeed, if a cycle existed, the sequence of triples

$$(e_i, e_{i+1}, e_{i+2}) \bmod 18$$

would form a directed cycle in G_3. By the lemma, the only possible directed cycles are the constant triples (a, a, a), which implies that all exponents along the cycle lie in the same residue class modulo 18.

In particular, the exponent sequence of any hypothetical odd Collatz cycle lies in a single residue class modulo 18.

A computational verification of Lemma 5.3 is given in Appendix A by explicit construction of G_3 and enumeration of its strongly connected components.

6 Refinement at Window Length $k = 4$

By Lemma 5.3, any exponent sequence arising from a hypothetical odd Collatz cycle satisfies

$$e_i \equiv r \pmod{18}$$

for some fixed residue $r \in \mathbb{Z}/18\mathbb{Z}$.

We therefore write

$$e_i = r + 18t_i, \qquad t_i \in \mathbb{Z}.$$

6.1 Reduction modulo 81

Specializing the window identity to length $k = 4$ and reducing modulo $3^4 = 81$, we analyze the resulting compatibility condition.

Since

$$e_i = r + 18t_i,$$

we have

$$2^{e_i} = 2^r (2^{18})^{t_i} \pmod{81}.$$

Now

$$2^{18} \equiv 28 \pmod{81}, \qquad 28^3 \equiv 1 \pmod{81},$$

so the residue of $(2^{18})^{t_i}$ modulo 81 depends only on t_i mod 3. Moreover, every power of 2 appearing in the length–four window identity is obtained from sums of exponents of the form

$$e_{i_1} + \cdots + e_{i_m} = mr + 18(t_{i_1} + \cdots + t_{i_m}),$$

and therefore also depends only on the corresponding residue classes t_{i_j} mod 3. Hence all powers of 2 occurring in the length–four residue map, and therefore the entire compatibility condition modulo 81, depend only on

$$t_i \in \mathbb{Z}/3\mathbb{Z}.$$

Define the residue map

$$R_4(a,b,c,d) = 2^{-(\epsilon(a)+\epsilon(b)+\epsilon(c)+\epsilon(d))}\Big(2^{\epsilon(d)+\epsilon(c)+\epsilon(b)} + 3\,2^{\epsilon(d)+\epsilon(c)} + 9\,2^{\epsilon(d)} + 27\Big) \pmod{81}, \tag{7}$$

where

$$\epsilon(x) = r + 18x, \qquad x \in \mathbb{Z}/3\mathbb{Z}.$$

6.2 Directed graph G_r

Define a directed graph G_r as follows:

- Vertices: all quadruples $(a,b,c,d) \in (\mathbb{Z}/3\mathbb{Z})^4$.
- Directed edge:

  $$(a,b,c,d) \longrightarrow (b,c,d,x)$$

  if and only if the compatibility condition

  $$R_4(b,c,d,x) \equiv \big(3R_4(a,b,c,d)+1\big)\,2^{-\epsilon(a)} \pmod{81}$$

  holds.

Lemma 6.1. *If an odd Collatz cycle exists with exponent residue $r \pmod{18}$, then the sequence of quadruples*

$$(t_i, t_{i+1}, t_{i+2}, t_{i+3})$$

forms a directed cycle in G_r.

Proof. As in the length–three case, every length–four window along the cycle, including those wrapping around the closing identification, must satisfy the window identity modulo 81. Since the window identity is an exact algebraic identity for every odd Collatz trajectory, the corresponding residue relation holds at every index of the cycle. Therefore consecutive windows satisfy the compatibility condition defining G_r, and hence produce a directed cycle. □

6.3 Elimination of nonconstant configurations

The key finite obstruction at window length four is the following.

Lemma 6.2. *For each $r \in \mathbb{Z}/18\mathbb{Z}$, the directed graph G_r has no directed cycles other than the constant ones*

$$(a,a,a,a) \to (a,a,a,a), \qquad a \in \mathbb{Z}/3\mathbb{Z}.$$

Lemma 6.2 is a finite statement about a directed graph with $3^4 = 81$ vertices. A computational verification is provided in Appendix A.

Consequently,

$$t_0 = t_1 = \cdots = t_{p-1},$$

and therefore

$$e_0 = e_1 = \cdots = e_{p-1}.$$

Hence any hypothetical odd Collatz cycle must have constant exponent sequence.

7 The Constant Exponent Case

By Section 6, any hypothetical odd Collatz cycle must have constant exponent sequence. Let

$$e_i = e \quad \text{for all } i.$$

7.1 Global cycle identity

Composing p consecutive odd steps yields the identity

$$n_{i+p} = \frac{2^{pe}}{3^p}\, n_i - \frac{1}{3^p}\sum_{j=0}^{p-1} 3^j\, 2^{(p-1-j)e}.$$

If the trajectory forms a cycle of length p, then $n_{i+p} = n_i$. Rearranging gives

$$n_i\big(2^{pe} - 3^p\big) = \sum_{j=0}^{p-1} 3^j\, 2^{(p-1-j)e}. \tag{8}$$

Solving for n_i yields

$$n_i = \frac{\displaystyle\sum_{j=0}^{p-1} 3^j 2^{(p-1-j)e}}{2^{pe} - 3^p}. \tag{9}$$

7.2 Case analysis

Case $e = 1$. Then $2^p - 3^p < 0$ for all $p \geq 1$, while the numerator in (9) is positive. Thus $n_i < 0$, which is impossible.

Case $e \geq 3$. Define the odd-to-odd map

$$T_e(n) = \frac{3n+1}{2^e}.$$

For $e \geq 3$ we have $2^e - 3 \geq 5$. For every $n \geq 1$,

$$3n + 1 < 2^e n \quad \iff \quad 1 < (2^e - 3)n.$$

Since $(2^e - 3)n \geq 5n \geq 5$, this inequality holds. Hence

$$T_e(n) < n.$$

Thus the odd terms strictly decrease at each step, which is incompatible with the existence of a cycle.

Case $e = 2$**.** In this case

$$T_2(n) = \frac{3n+1}{4}.$$

Equation (9) yields $n_i = 1$ for all $p \geq 1$. This corresponds to the trivial odd cycle.

7.3 Conclusion

The analysis above shows that if the exponent sequence of an odd Collatz cycle is constant, then the only possible cycle is the trivial one corresponding to $n = 1$.

Proof of the Main Theorem. Assume a nontrivial odd Collatz cycle exists.

By Section 5, the exponent sequence must be constant modulo 18. By Section 6, it must therefore be constant. Section 7 shows that the constant exponent case produces only the trivial cycle.

This contradicts the assumption that a nontrivial odd Collatz cycle exists. □

Conclusion

We have shown that the existence of an odd Collatz cycle would force the associated exponent sequence to satisfy strong local compatibility conditions arising from the window identities. Encoding these conditions as finite directed graphs yields a complete finite obstruction, forcing the exponent sequence to be constant. The constant case produces only the trivial cycle.

Hence no nontrivial odd Collatz cycles exist.

Appendix A Computational Verification of the Graph Constraints

The statements of Lemma 5.3 and Lemma 6.2 reduce to finite computations on directed graphs.

A.1 Construction of G_3

The graph G_3 has vertex set

$$(\mathbb{Z}/18\mathbb{Z})^3,$$

so it contains $18^3 = 5832$ vertices.

For each vertex (a, b, c) and each $d \in \mathbb{Z}/18\mathbb{Z}$, we test whether the compatibility condition

$$R_3(b, c, d) \equiv (3R_3(a, b, c) + 1)\, 2^{-a} \pmod{27}$$

holds. If so, we add the directed edge

$$(a, b, c) \to (b, c, d).$$

The resulting directed graph has 5832 vertices and 5832 edges. In particular, each vertex has out-degree exactly one.

A.2 Cycle detection

To detect cycles, we compute the strongly connected components (SCCs) of the directed graph using Tarjan's algorithm. The computation shows that the only SCCs containing directed cycles are the trivial self-loops

$$(a, a, a) \to (a, a, a)$$

for $a \in \mathbb{Z}/18\mathbb{Z}$.

In particular, G_3 contains no nonconstant directed cycles.

A.3 Verification for $k = 4$

For the refinement step, the graph G_r has vertex set

$$(\mathbb{Z}/3\mathbb{Z})^4,$$

containing $3^4 = 81$ vertices. This verification is carried out separately for each residue

$$r \in \mathbb{Z}/18\mathbb{Z},$$

so in total 18 directed graphs are checked. Edges are defined by the compatibility condition modulo 81.

The same SCC analysis shows that the only directed cycles are the constant ones

$$(a, a, a, a) \to (a, a, a, a), \qquad a \in \mathbb{Z}/3\mathbb{Z}.$$

A.4 Implementation

The computations were implemented in Python. The source code used to construct the graphs and enumerate their strongly connected components is publicly available at
`https://github.com/jmarchionatto/collatz-3adic-cycle-obstruction`.

AI Assistance Disclosure

This work was developed with assistance from large language models for drafting, structuring, and refining exposition, as well as for checking algebraic derivations. The mathematical ideas, results, and overall research direction are due to the author, who takes full responsibility for the content.

References

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